

p53: guardian of the genome and policeman of the oncogenes.

The process of malignant transformation universally entails genetic damage and oncogenic signaling, two stresses that are signaled to p53 through different genetic pathways. Based on this, it is possible to distinguish two jobs for p53: "guardian of the genome" that consists in sensing and reacting to DNA damage through the ATM/ATR and Chk1/Chk2 kinases, and "policeman of the oncogenes" that, correspondingly, consists in responding to oncogenic signaling through the p53-stabilizing protein ARF. Contrary to expectation, recent genetic evidence in mice indicates that the response of p53 to DNA damage has little or no impact on cancer protection. In contrast, ARF-dependent activation of p53 is critical for p53-mediated tumor suppression. Here, we discuss the mechanistic implications of these observations and their relevance for cancer therapy.